DIGITAL IMAGE PROCESSING

LECTURE # 7

IMAGE ENHANCEMENT IN SPATIAL DOMAIN-IV

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Topics to Cover

- Gradient Operators
  - **Robert Cross Operator**
  - **Prewitt Operator**
  - **Sobel Operator**
  - **Robinson Compass Operator**

1st Derivative Filtering- The Gradient

- 1st derivative filters is Gradient which represents change
- For a function **f(x,y)** the gradient of f at coordinates **(x,y)** is given as the column vector:

$$\nabla f = \begin{bmatrix} G_x \\ G_y \end{bmatrix} = \begin{bmatrix} \dfrac{\partial f}{\partial x} \\ \dfrac{\partial f}{\partial y} \end{bmatrix}$$

1st Derivative Filtering- The Gradient

- The magnitude of this vector is given by:

$$\nabla f = mag(\nabla \mathrm{f})$$
$$= \left[G_x^2 + G_y^2\right]^{1/2}$$
$$= \left[\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2\right]^{1/2}$$

- The direction of this vector is given by:

$$\theta = \tan^{-1}\left[G_y / G_x\right]$$

Roberts Cross Operator

- Two other definitions proposed by Roberts use cross- difference

$$G_x = (z_9 - z_5) \quad \text{and} \quad G_y = (z_8 - z_6).$$

- If we use

$$\nabla f = \left[\left(\frac{\partial f}{\partial x} \right)^2 + \left(\frac{\partial f}{\partial y} \right)^2 \right]^{1/2}$$

$$\nabla f = \left[(z_9 - z_5)^2 + (z_8 - z_6)^2 \right]^{1/2}$$

-1	0
0	1

0	-1
1	0

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

Roberts Cross-Gradient Operators

Prewitt & Sobel Operator

Prewitt & Sobel operator is used for edge detection in an image.

It detects two types of edges

- Vertical edges
- Horizontal Edges

Prewitt Operator

Vertical direction

-1	0	1
-1	0	1
-1	0	1

- This mask will find the edges in vertical direction and it is because the zeros column in the vertical direction

- When we apply this mask on the image it prominent vertical edges
- It simply works like as first order derivate and calculates the difference of pixel intensities in a edge region
- As the center column is of zero so it does not include the original values of an image but rather it calculates the difference of right and left pixel values around that edge.
- This increase the edge intensity and it become enhanced comparatively to the original image.

Prewitt Operator

Horizontal Direction

-1	-1	-1
0	0	0
1	1	1

- This mask will find edges in horizontal direction and it is because that zeros column is in horizontal direction.
- When you will convolve this mask onto an image it would prominent horizontal edges in the image.
- This mask will prominent the horizontal edges in an image
- It also works on the principle of above mask and calculates difference among the pixel intensities of a particular edge.
- As the center row of mask is consist of zeros so it does not include the original values of edge in the image but rather it calculate the difference of above and below pixel intensities of the particular edge.
- Thus increasing the sudden change of intensities and making the edge more visible.

Prewitt Operator Combine

- Normally the smallest mask used is of size 3 x 3
- Based on the concept of approximating the gradient several spatial masks have been proposed

Prewitt operator

$$\begin{array}{c|ccc} & & y & \\ \hline x & -1 & -1 & -1 \\ & 0 & 0 & 0 \\ & 1 & 1 & 1 \end{array}$$

Extract horizontal edges

$$\begin{array}{c|ccc} & & y & \\ \hline x & -1 & 0 & 1 \\ & -1 & 0 & 1 \\ & -1 & 0 & 1 \end{array}$$

Extract vertical edges

Pixel arrangement

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

Prewitt operators (equation):

$$\nabla f \approx \left|(z_7 + z_8 + z_9) - (z_1 + z_2 + z_3)\right| + \left|(z_3 + z_6 + z_9) - (z_1 + z_4 + z_7)\right|$$

Sobel Operator

-1	0	1
-2	0	2
-1	0	1

Vertical direction

- This mask works exactly same as the Prewitt operator vertical mask.
- There is only one difference that is it has “2” and “-2” values in center of first and third column.
- When applied on an image this mask will highlight the vertical edges
- As the center column is of zero so it does not include the original values of an image but rather it calculates the difference of right and left pixel values around that edge.
- Also the center values of both the first and third column is 2 and -2 respectively.

Sobel Operator

Horizontal Direction

-1	-2	-1
0	0	0
1	2	1

- This mask will find edges in horizontal direction and it is because that zeros column is in horizontal direction.
- When you will convolve this mask onto an image it would prominent horizontal edges in the image.
- This mask will find edges in horizontal direction and it is because that zeros column is in horizontal direction.
- When you will convolve this mask onto an image it would prominent horizontal edges in the image.
- The only difference between it is that it have 2 and -2 as a center element of first and third row

Sobel Operator Combine

Sobel operators

Extract horizontal edges (x down, y across):

$$\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

Extract vertical edges (x down, y across):

$$\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

Sobel operators (equation):

Emphasize more the current position (y)

$$\nabla f \approx \left|(z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3)\right| + \left|(z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7)\right|$$

Emphasize more the current position (x)

Pixel arrangement

$$\begin{matrix} z_1 & z_2 & z_3 \\ z_4 & z_5 & z_6 \\ z_7 & z_8 & z_9 \end{matrix}$$

Robinson Compass Operator

- Robinson compass masks are another type of derivate mask which is used for edge detection.
- This operator is also known as direction mask.
- In this operator we take one mask and rotate it in all the 8 compass major directions that are following:
  - North
  - North West
  - West
  - South West
  - South
  - South East
  - East
  - North East
- There is no fixed mask.
- You can take any mask and you have to rotate it to find edges in all the above mentioned directions.
- All the masks are rotated on the bases of direction of zero columns.

Robinson Compass Operator

North West

0	1	2
-1	0	1
-2	-1	0

North

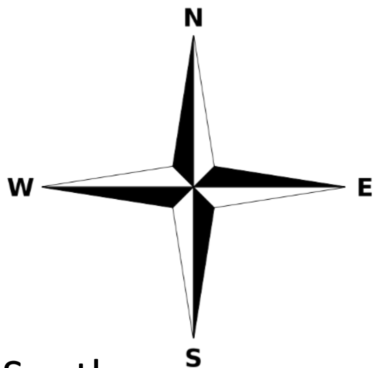
-1	0	1
-2	0	2
-1	0	1

North East

-2	-1	0
-1	0	1
0	1	2

West

1	2	1
0	0	0
-1	-2	-1



East

-1	-2	-1
0	0	0
1	2	1

South West

2	1	0
1	0	-1
0	-1	-2

South

1	0	-1
2	0	-2
1	0	-1

South East

0	-1	-2
1	0	-1
2	1	0

Robinson Compass Operator

What will be the effect of if compare filter with/without absolute in the equation?

- North vs south
- West vs east
- North west vs south east
- North east vs south west

Derivatives Operators Observation

All the derivative masks should have the following properties:

- Opposite sign should be present in the mask.
- Sum of mask should be equal to zero.
- More weight means more edge detection.

Class Task

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References

1. DIP by Gonzalez

For any query Feel Free to ask